

Sungjun Lee

Vancouver, BC | 778-926-8205 | anderpuding0917@gmail.com | GitHub | Portfolio

Education

University of British Columbia

Expected May 2027

Bachelor of Science, Combined Major in Computer Science and Mathematics

Vancouver, BC

Coursework: Data Structures and Algorithms, Software Construction, Probability, Optimization, Linear Algebra, Differential Equations

Technical Skills

Languages: Python, Java, Swift, TypeScript, JavaScript, SQL, C, C++, MATLAB

Frameworks and Cloud: React, Next.js, Node.js, Express, SwiftUI, Supabase, PostgreSQL, Firebase, Docker

AI and Data: OpenAI API, Gemini API, NumPy, pandas, SciPy, scikit-learn, TensorFlow, Matplotlib, Plotly

Tools: Git, GitHub, Linux, Jupyter, VS Code, Xcode, Supabase CLI, REST APIs

Experience

Software Engineer Intern – AI/ML

Apr. 2026–Aug. 2026

Product Manager Accelerator

Remote

Project: Remember – AI-Powered Memorial Platform

- Delivered a **full-stack** memorial minimum viable product (MVP) to 35+ internal and external testers by building new pages, modifying ~50 existing pages, and taking ownership of the end-to-end contributor workflow.
- Expanded frontend-backend coverage to 26 active **API endpoints** by implementing 17 integrations for onboarding, autosave, media uploads, submission, organizer review, and AI-generation status tracking.
- Optimized media-upload and **data-processing workflows**, reducing persistence latency by 70% for text, image, audio, and video submissions and accelerating AI-generated response delivery.
- Accelerated delivery across a team of five developers and two data scientists by contributing 100+ **commits**
- Authored 17+ **merged pull requests** for a Vercel and Render-deployed Next.js, Express, Supabase, PostgreSQL, and OpenAI application.

Project: WonderWordAI – AI Reading Coach Platform

- Built a full-stack reading-coach MVP across a four-week sprint by owning the Next.js web shell, Supabase Auth, PostgreSQL schema, **Row Level Security policies**, and Vercel deployment setup.
- Implemented 20 **Next.js API routes** and end-to-end session workflows for worksheet capture, OCR forwarding, audio submission, parent dashboard reports, story narration, phonics lookup, and practice recommendations.
- Hardened release readiness by adding **Playwright E2E coverage**, deterministic test/typecheck workflows, 15 Supabase migrations, and pgTAP coverage for auth, RLS, session access, and known-word refresh behavior.

Research Assistant – Planning for Climate Equity

Vancouver, BC

PLACE Lab, University of British Columbia

- Improved the reproducibility of climate-equity analysis by building reusable **Python workflows** with pandas, Matplotlib, and Plotly that cleaned, joined, modeled, and visualized climate and demographic data.
- Developed **data analysis and visualization workflows** that helped researchers interpret complex datasets and streamline research efforts.

Technical Projects

Vanori – AI-Powered Local Discovery iOS Application

May. 2026–Jun. 2026

Swift, SwiftUI, Supabase, PostgreSQL, Edge Functions, Gemini

- Shipped a TestFlight **MVP** with approximately 40 screens and 100 curated places, events, and activities by building authentication, onboarding, discovery, saved-item, review, notification, and profile workflows in SwiftUI.
- Improved recommendation relevance and explainability for cold-start users by designing a **hybrid recommendation system** that scores content across seven weighted factors, filters the top 20 candidates, and uses Gemini 2.5 Flash-Lite to rerank the top five with user-facing recommendation reasons.
- Reduced **hallucination** and data-quality risk in AI recommendations by limiting Gemini to validated candidate items and structuring the backend around five core data models, seven content sources, PostgreSQL, Row Level Security, Supabase Storage, and Edge Functions.

AlgoQuantEngine – Quantitative Portfolio Optimization System

Python

- Built a reproducible portfolio-analysis engine to help users compare investment strategies beyond raw return, as measured by support for **synthetic universes** of up to 120 assets and seven modules covering data processing, optimization, graph clustering, Monte Carlo risk, backtesting, reporting, and benchmarking.
- Increased trust in quantitative results by validating 19 of 19 **automated tests** across nine test files and benchmarking mean-variance portfolios against equal-weight and minimum-variance strategies with CSV, JSON, PNG, and command-line reporting outputs.